CHARLIE'S ANGELS
Capstone Project
Migration & Stabilization Team Challenge
Auree-Lynn | Christina | Charles | Aijahnae
... AND JOHN
PLAN
MIGRATE
TROUBLESHOOT
STABILIZE
IMPROVE
PER SCHOLAS
AI-Enabled IT Support

# PROJECT OVERVIEW

01 V-LAN Setup

02 Enterprise Network Setup

03 Migration Event

04 Ticket Breakdown

05 Executive Summary

# VIRTUAL LAN SETUP

| | SERVER VM | CLIENT VM |
|---|---|---|
| OS: | WINDOWS SERVER 2022 | WINDOWS 11 |
| VM NAME: | MIG-SRV01 | MIG-CL01 |
| CPUS: | 2 CORES | 2 CORES |
| RAM: | 6 GB | 4 GB |
| STORAGE: | 80 GB | 60 GB |
| HYPERVISOR: | VIRTUALBOX | VIRTUALBOX |
| NETWORK ADAPTER: | INTERNAL NETWORK | INTERNAL NETWORK |
| NETWORK NAME: | SAME ON BOTH VMS | SAME ON BOTH VMS |

# ENTERPRISE NETWORK CONFIGURATION

## DOMAIN CONFIGURATION

- Domain (FQDN): **migration.local**
- NetBIOS: **MIGRATION**
- Active Directory Domain Services (AD DS)
- DNS for domain name resolution
- Windows Server 2022 configured as Domain Controller

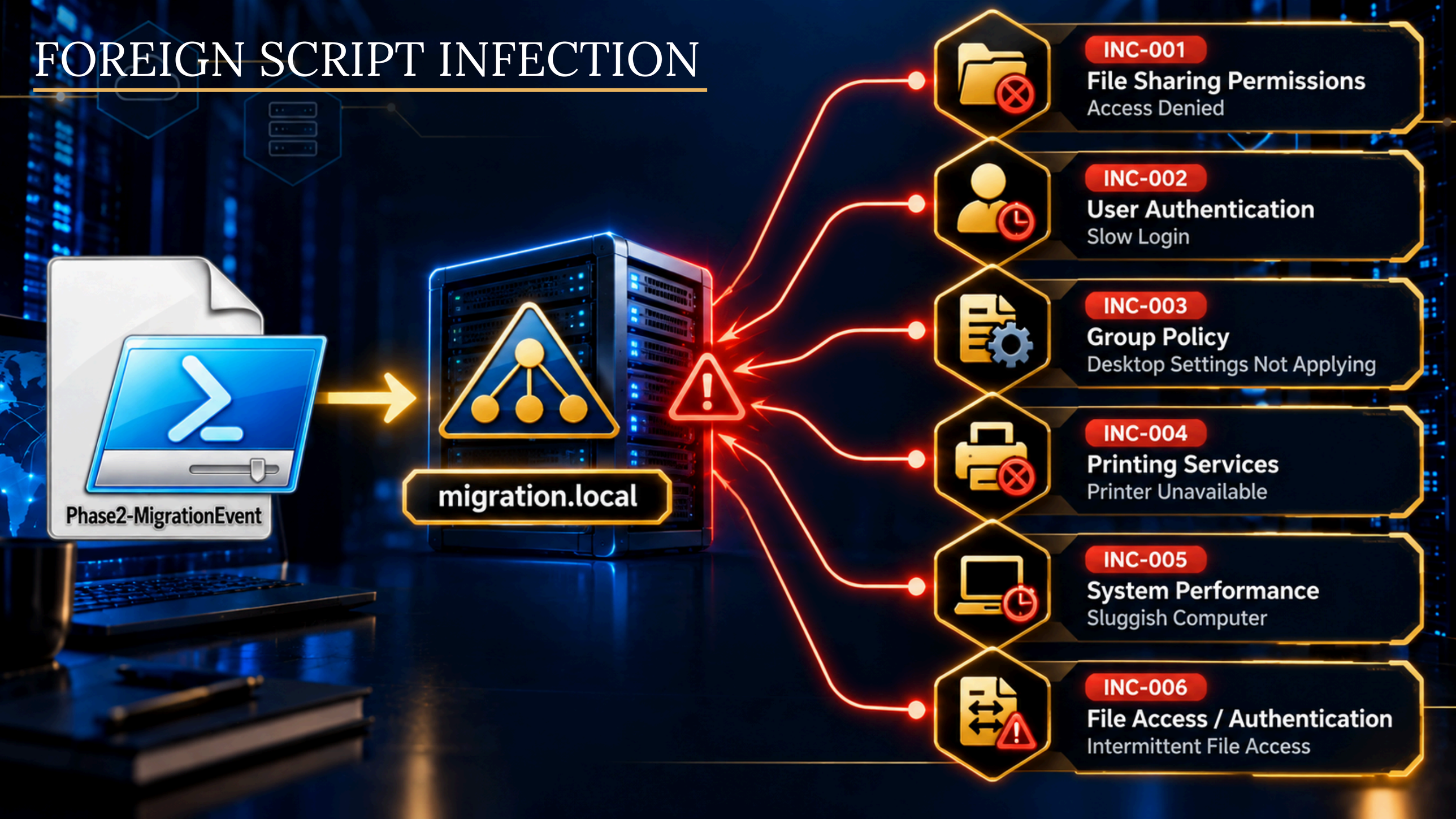
## IDENTITY & ACCESS MANAGEMENT

- Domain Administrator: **MigrationAdmin**
- Standard User: **MigrationUser**
- Administrator Group: **IT-Admins**
- Standard User Group: **Domain-Users**
- Windows 11 client joined to **migration.local** domain

## SHARED NETWORK RESOURCES

- Shared Folder Path: **C:\SharedData**
- SMB Share Name: **SharedData**
- Security Group: **FileShareUsers**
- Access: Authorized domain users can access the shared folder through group permissions

# FOREIGN SCRIPT INFECTION



# INC-001 | UNABLE TO ACCESS SHARED FILES

PHASE 3 • ISSUE & INVESTIGATION

## USER-REPORTED ISSUE

Finance user reported “Access Denied” when attempting to open the company SharedData folder.

## INVESTIGATION STEPS

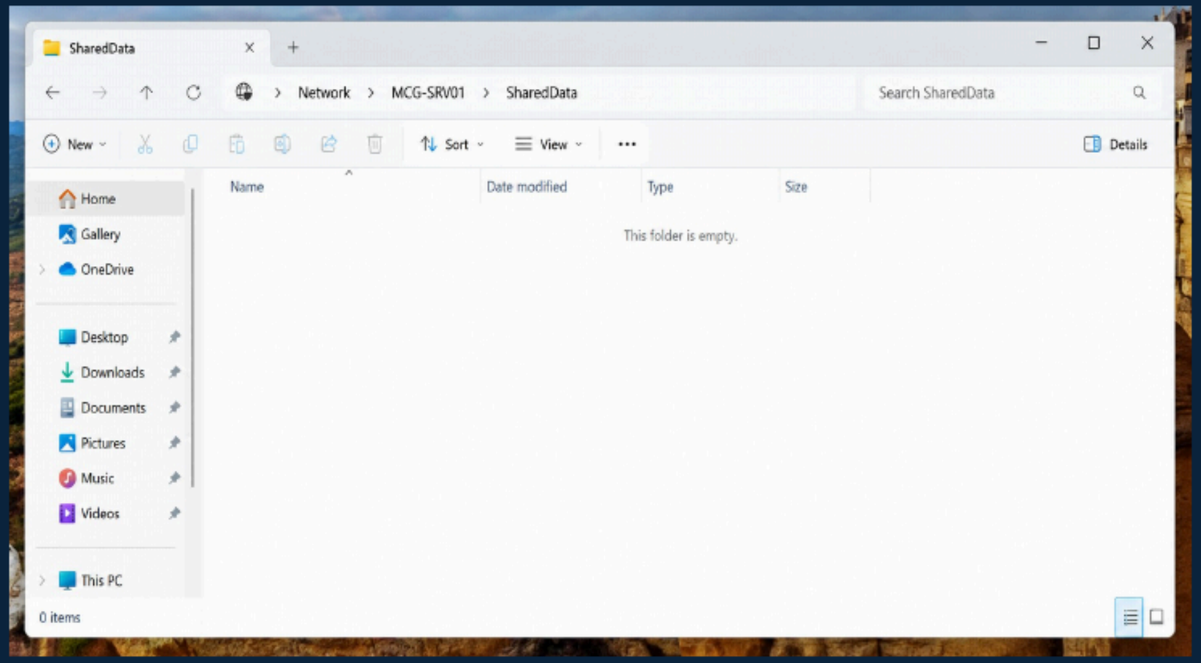
01 Tested connectivity to server IP and hostname

02 Checked domain identity and group membership

03 Attempted access to the shared folder over SMB

04 Reviewed share-level and NTFS access permissions

## ACTUAL EVIDENCE • SHARED RESOURCE




**File Explorer shows the SharedData network share opening.**

This supports access verification, not the earlier Access Denied event or the specific permission changes.

# INC-001 | ACTIVE DIRECTORY INVESTIGATION

PHASE 3 • EVIDENCE REVIEW

## WHAT WE CHECKED

01 **Domain identity**

Reviewed the migration.local directory.

02 **Security groups**

Located the FileShareUsers group.

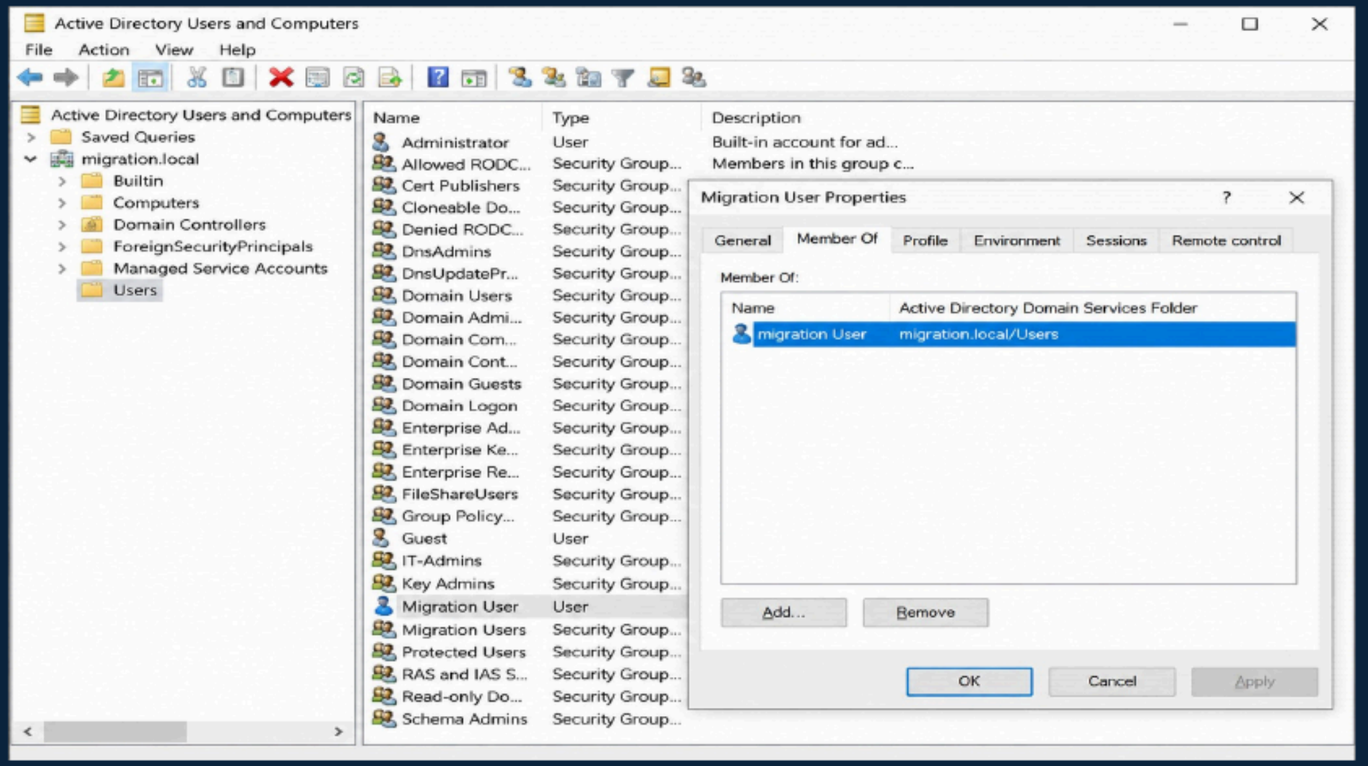
03 **User membership**

Inspected the account's "Member Of" tab.

04 **Access diagnosis**

Group access needed further verification.

## ACTUAL EVIDENCE • AD USERS AND COMPUTERS



# INC-001 | RESOLUTION & VERIFICATION

## UNABLE TO ACCESS SHARED FILES

### GROUP-REPORTED SOLUTION

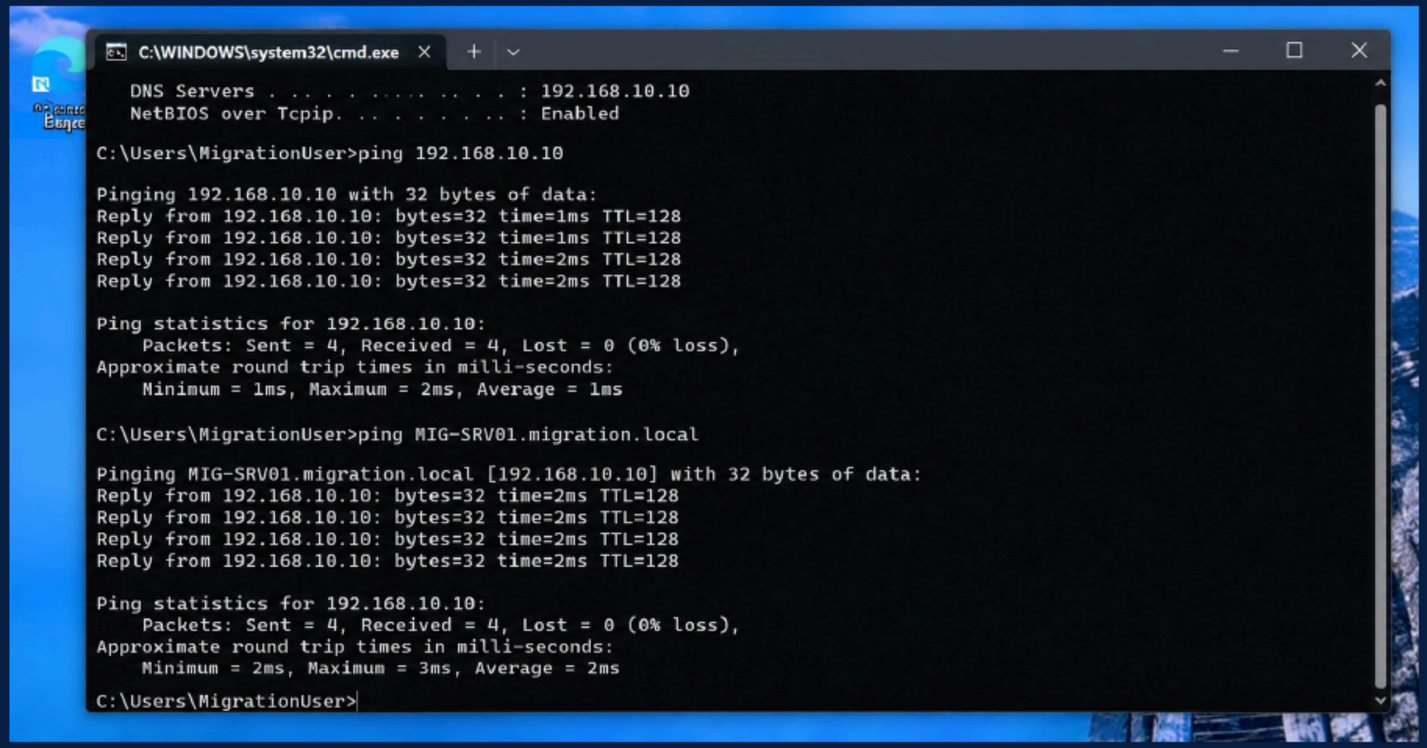
01 Reviewed share and NTFS access controls

02 Corrected access for the FileShareUsers security group

03 Confirmed domain user access to SharedData

**Reported result: access restored**

### ACTUAL CONNECTIVITY TEST



# INC-002 | SLOW DOMAIN LOGIN

## ISSUE & INVESTIGATION

### REPORTED BY

**Mary Jones | Human Resources**

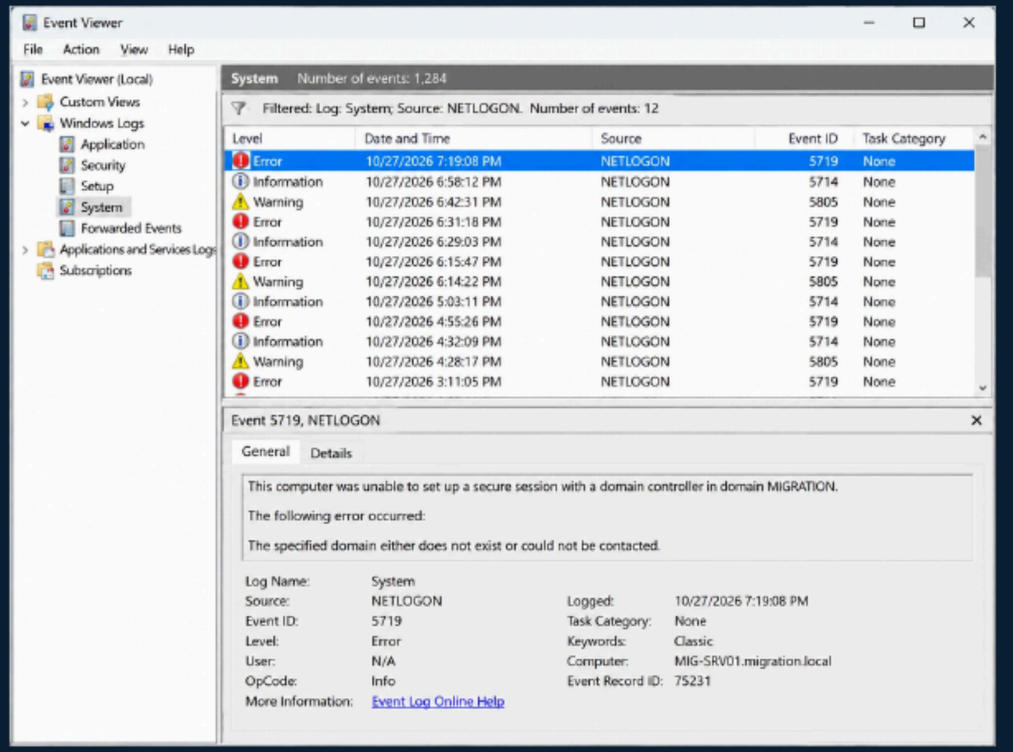
### USER-REPORTED ISSUE

Unusually slow domain sign-in following the migration, impacting productivity.

### TROUBLESHOOTING PERFORMED

- Tested connection to 192.168.10.10
- Reviewed Event Viewer: NETLOGON 5719
- Investigated Netlogon and authentication
- Checked whether delay was reproducible

### EVENT VIEWER — NETLOGON EVENT 5719




**Verification Performed**

Ping to 192.168.10.10 failed
4 packets lost — 100% packet loss

# INC-002 | SLOW DOMAIN LOGIN

PHASE 4 • RESOLUTION & VERIFICATION

## REMEDIATION PERFORMED

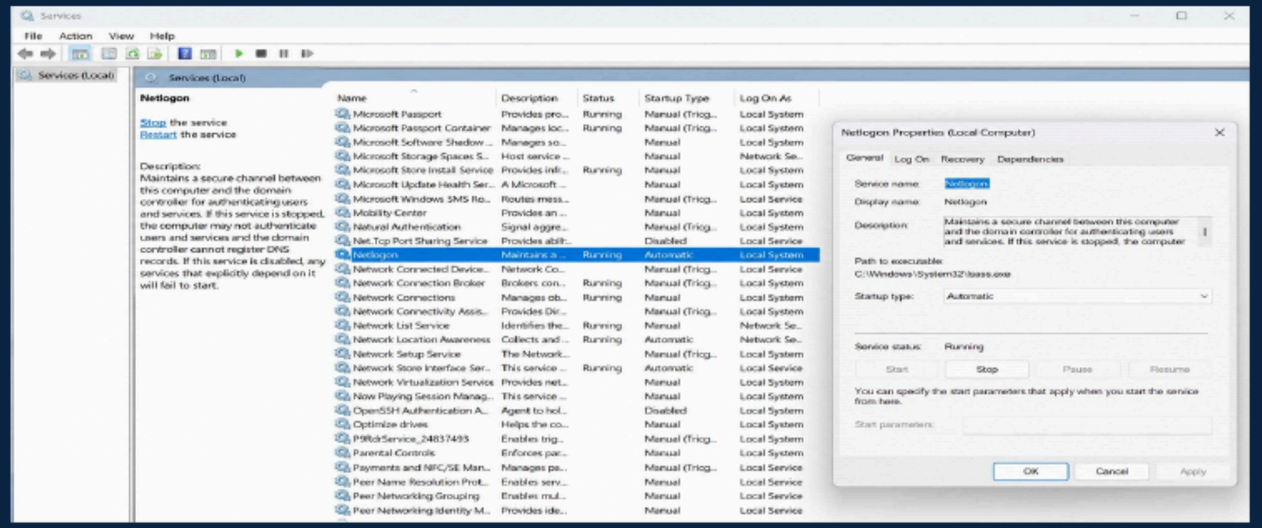
01 Verified DNS and domain controller connectivity.

02 Inspected the Netlogon service in Windows Services.

03 Confirmed Automatic startup configuration.

04 Restarted Netlogon to restore domain communication.

## ACTUAL EVIDENCE • NETLOGON SERVICE




**Services shows Netlogon: Running | Startup: Automatic**

## VERIFICATION RESULTS

- Service: Running / Automatic
- Client communication restored (team-reported)
- Domain login retested with MIGRATION\MigrationUser
- Login under 5 seconds (team-reported)

# INC-003 | DESKTOP SETTINGS NOT APPLYING

## ISSUE & INVESTIGATION

**REPORTED BY**

**John Smith | Finance**

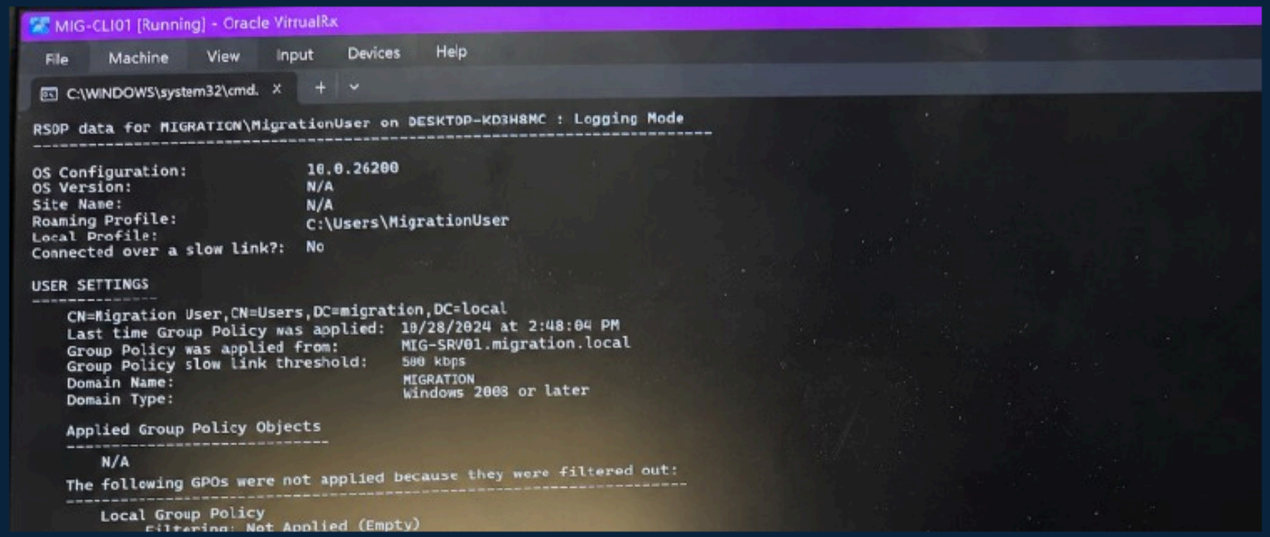
### USER-REPORTED ISSUE

Normal desktop settings appeared to be missing after signing in to the domain.

### TROUBLESHOOTING PERFORMED

- Confirmed communication with MIG-SRV01
- Ran gpresult /r to inspect applied GPOs
- Reviewed Default Domain Policy in GPMC
- Investigated SYSVOL and AD permissions

### GROUP POLICY RESULTS — gpresult /r




**Applied Group Policy Objects: N/A**

**KEY FINDINGS** Default Domain Policy linked/enabled; team reported GPMC SYSVOL–AD permission inconsistency.

# INC-003 | DESKTOP SETTINGS NOT APPLYING

**PHASE 4 • RESOLUTION & VERIFICATION**

## REMEDIATION PERFORMED

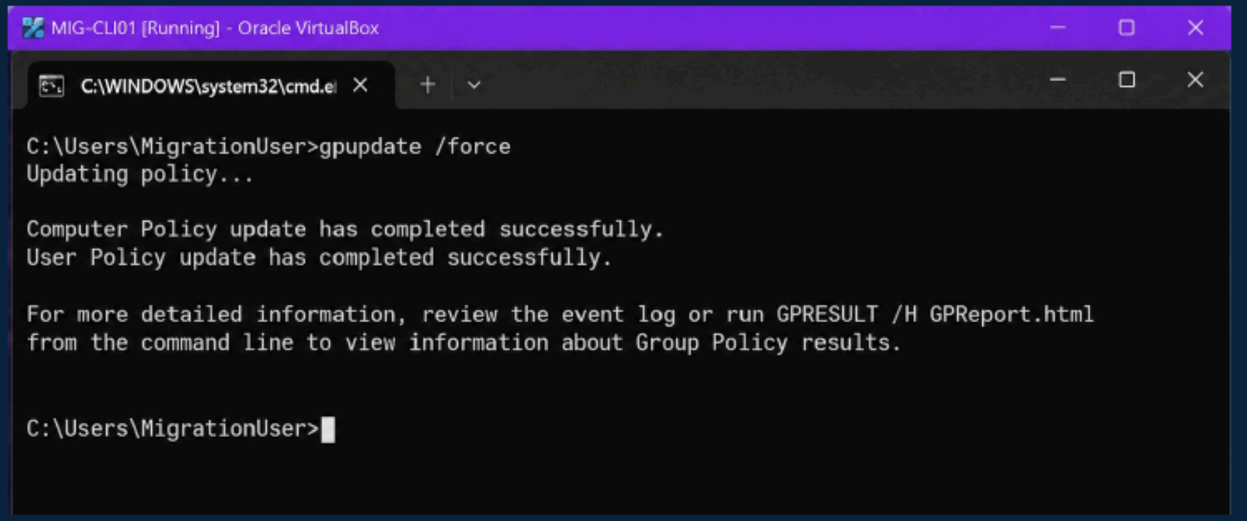
01 Verified Default Domain Policy was enabled and linked.

02 Reviewed GPO permissions and security filtering.

03 Investigated SYSVOL / Active Directory permission mismatch.

04 Ran gpupdate /force to refresh computer and user policy.

## ACTUAL EVIDENCE • GROUP POLICY REFRESH




**gpupdate /force: computer and user policy updates completed successfully.**

## VERIFICATION RESULTS

- Group Policy refresh completed successfully (shown above)
- Domain sign-in and gpresult /r retested (team-reported)
- Expected desktop settings restored (team-reported)

# INC-004 | PRINTER NO LONGER AVAILABLE

## ISSUE & INVESTIGATION

**REPORTED BY**

**Mary Jones | Human Resources**

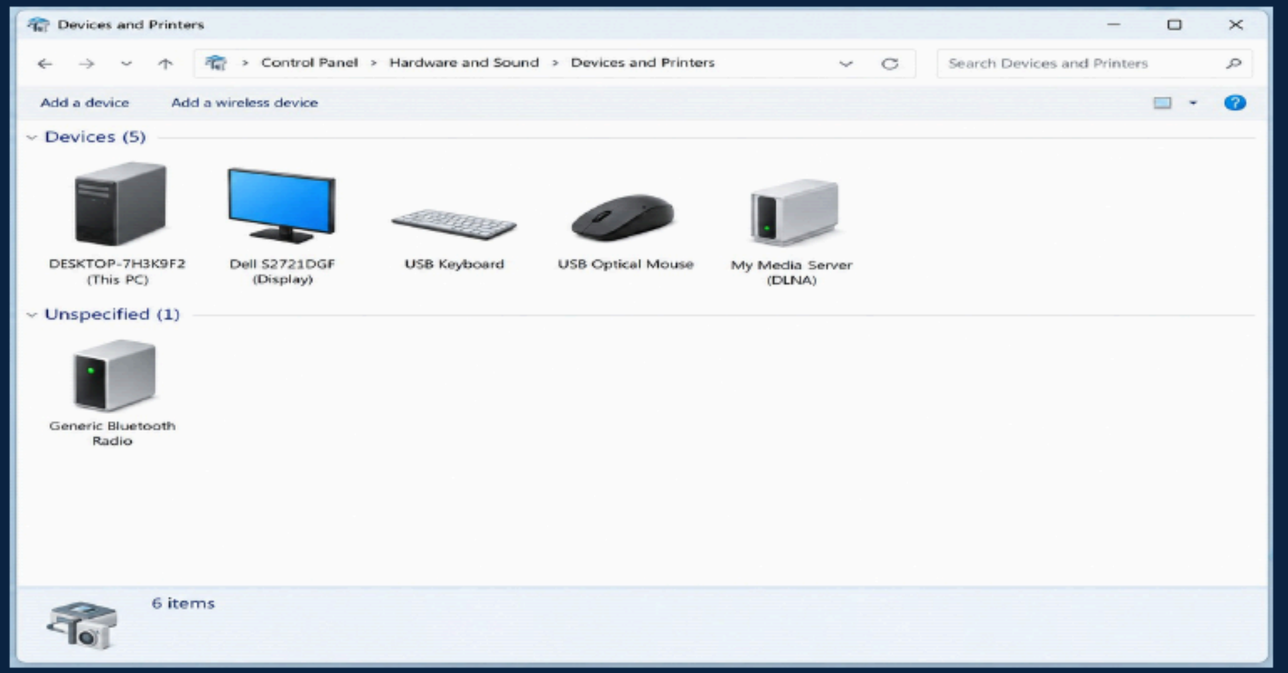
### USER-REPORTED ISSUE

The usual network printer was missing after the migration, preventing completion of printed documents.

### TROUBLESHOOTING PERFORMED

- Checked printer availability in Control Panel
- Reviewed Print Spooler service status
- Investigated server printer configuration and sharing
- Checked whether the expected network printer was available

### ACTUAL EVIDENCE • DEVICES AND PRINTERS




**Expected network printer not listed on the client.**

**KEY FINDINGS** Printer absent on client; Print Spooler running and server share missing (team-reported).

# INC-004 | PRINTER NO LONGER AVAILABLE

**PHASE 4 • RESOLUTION & VERIFICATION**

## REMEDIATION PERFORMED

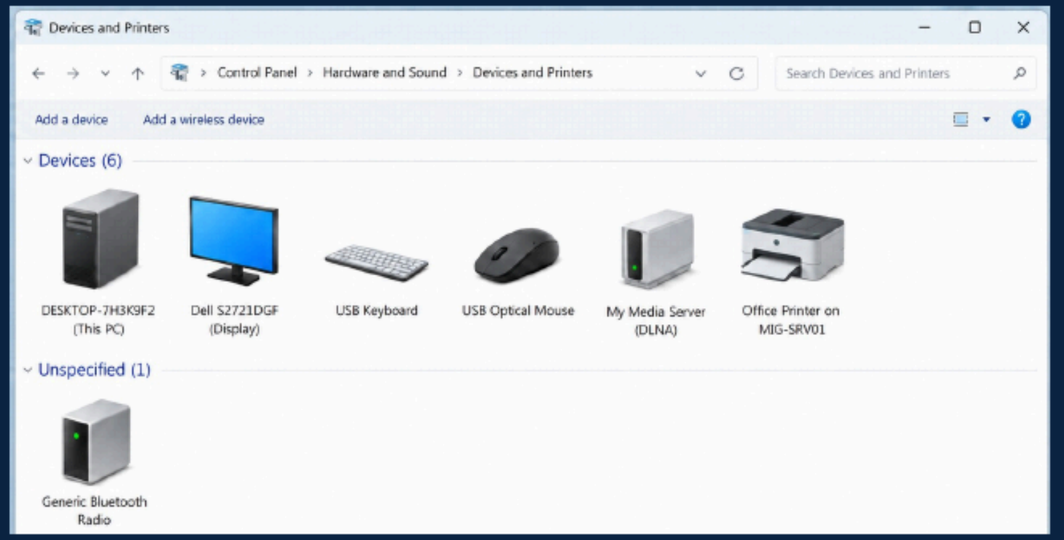
01 Verified Print Spooler was enabled and set to Automatic.

02 Restarted Print Spooler to address possible service issues.

03 Reviewed printer installation and sharing configuration.

04 Added or reconnected the network printer on the client.

## ACTUAL EVIDENCE • PRINTER RESTORED




**Devices and Printers shows "Office Printer on MIG-SRV01" listed on the client.**

## VERIFICATION RESULTS

Printer visible on client (shown above) • Print Spooler running (team-reported)
Printer available for use; successful test print (team-reported)

# INC-005 | SYSTEM FEELS SLUGGISH

## ISSUE & INVESTIGATION

**REPORTED BY**

**John Smith | Finance**

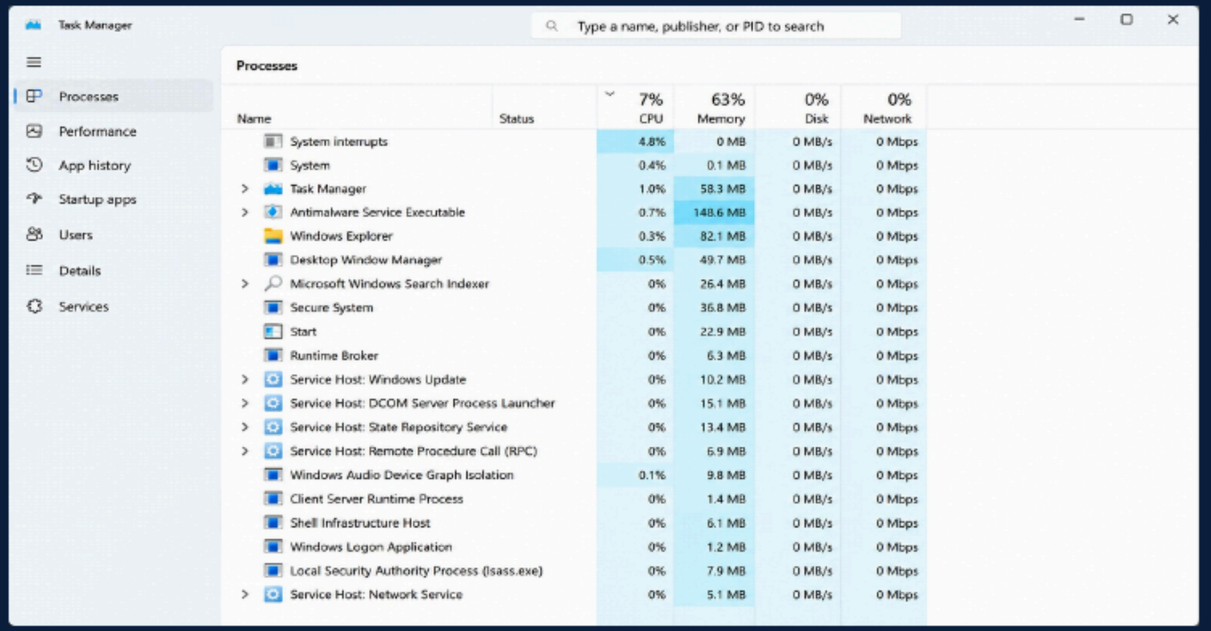
### USER-REPORTED ISSUE

Applications opened slowly and the workstation felt less responsive following the migration.

### TROUBLESHOOTING PERFORMED

- Reviewed CPU, memory, disk and network use in Task Manager
- Checked Task Scheduler for unexpected background activity
- Investigated the EnterpriseMaintenanceTask scheduled task

### ACTUAL EVIDENCE • TASK MANAGER




**7% CPU | 63% MEMORY | 0% DISK | 0% NETWORK**

**KEY FINDINGS** No sustained spike observed; background task investigated (team notes).

# INC-005 | SYSTEM FEELS SLUGGISH

**PHASE 4 • PERFORMANCE TROUBLESHOOTING & OPTIMIZATION**

## REMEDIATION & FOLLOW-UP

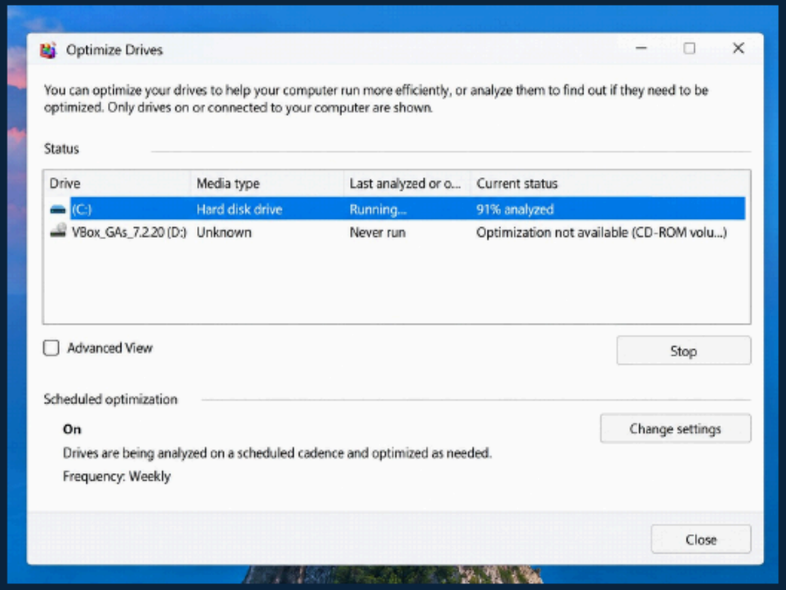
01 Stopped EnterpriseMaintenanceTask and retested responsiveness.

02 Slowdown persisted after disabling the background task.

03 Investigated HDD-backed client VM storage as a possible limitation.

04 Reviewed drive analysis and optimization as a next step.

## ACTUAL EVIDENCE • OPTIMIZE DRIVES




**C: identified as HDD • drive analysis in progress (91%).**

## CURRENT STATUS

Scheduled-task change did not resolve the slowdown. Drive optimization is being evaluated; performance recovery is not yet verified.

Evidence note: Screenshot shows HDD classification and analysis in progress; it does not demonstrate improved performance.

# INC-006 | INTERMITTENT FILE ACCESS ISSUES

## ISSUE & INVESTIGATION

**REPORTED BY**

**Mary Jones | Human Resources**

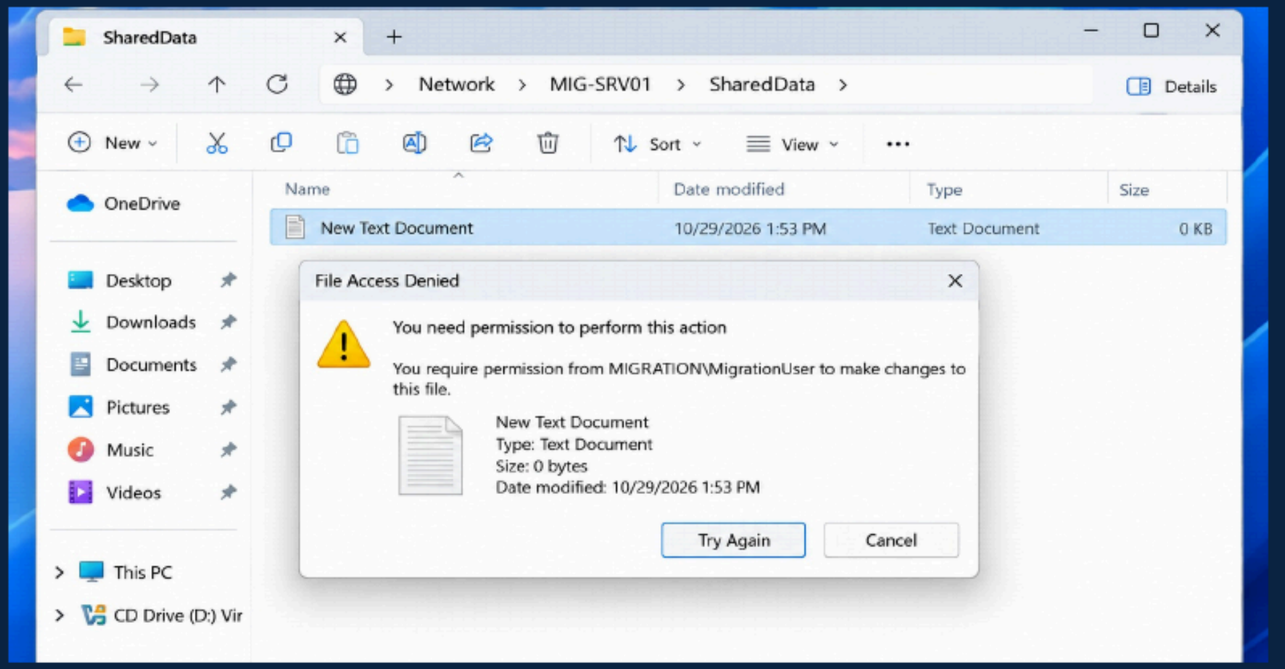
### USER-REPORTED ISSUE

Intermittent access to \\MIG-SRV01\\SharedData disrupted access to shared company resources.

### TROUBLESHOOTING PERFORMED

- Tested the shared-folder UNC path on the Windows 11 client
- Checked connectivity to MIG-SRV01 (192.168.10.10)
- Checked hostname resolution and internal DNS
- Investigated SMB share and NTFS permissions

### ACTUAL EVIDENCE • SHARED-FILE ACCESS




**File Access Denied while attempting to modify a file on the share.**

### KEY FINDINGS

Share was reachable; file modification was denied. Team also reported successful hostname resolution and a failed DNS recursive query.

# INC-006 | INTERMITTENT FILE ACCESS ISSUES

## PHASE 4 • RESOLUTION & VERIFICATION

### REMEDIATION PERFORMED

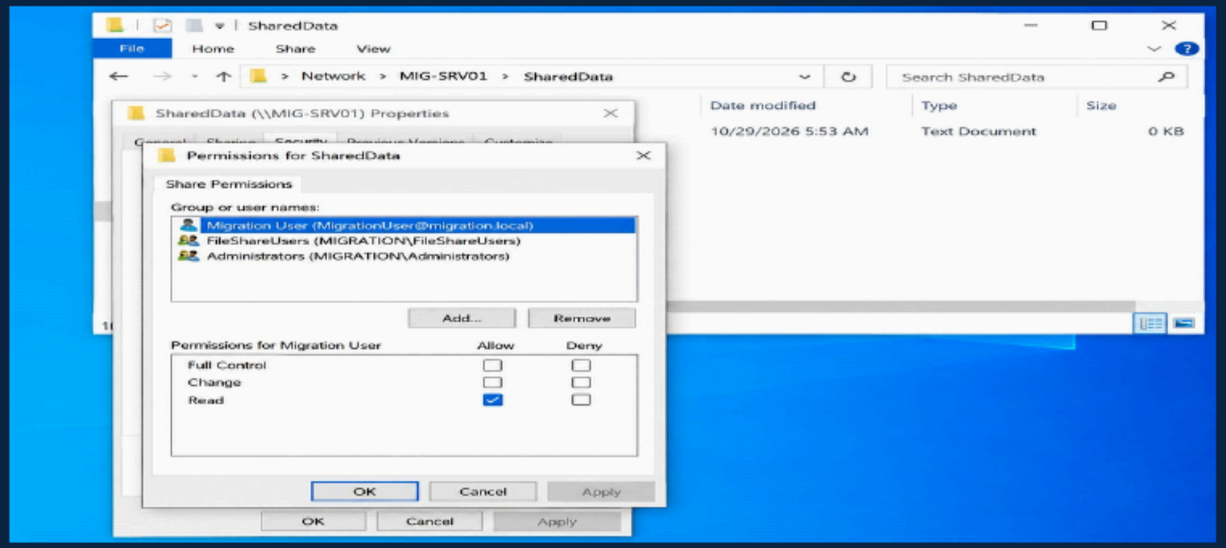
01 Opened folder security settings for C:\SharedData.

02 Reviewed and updated permissions for authorized access.

03 Checked required domain users and security groups.

04 Applied the updated settings.

### ACTUAL EVIDENCE • SHARE PERMISSIONS




**Screenshot: Migration User has Read allowed; Change not checked.**

### VERIFICATION RESULTS

- Retested \\MIG-SRV01\SharedData from Windows 11
- Authorized domain access restored (team-reported)
- Communication with MIG-SRV01 verified
- Intermittent access issue resolved (team-reported)

Reported outcome: Shared-folder access restored after permission corrections.

# EXECUTIVE SUMMARY

## CHARLIE'S ANGELS

Migration & Stabilization Capstone

**OUR RESPONSIBILITY**

Maintain and restore enterprise IT services following a system migration.

**THE ENVIRONMENT**

Windows Server 2022 domain controller and Windows 11 client in a virtualized network.

**OUR RESPONSE**

Investigated six support incidents, applied corrective actions where appropriate, and tested the results.

Core services: authentication • file sharing • desktop configuration • printing

# POST-MIGRATION ISSUES & BUSINESS IMPACT

EXECUTIVE SUMMARY | CHARLIE'S ANGELS

## SIX REPORTED INCIDENTS

01 Shared company files inaccessible

02 Slow domain sign-in

03 Missing desktop settings

04 Network printer unavailable

05 Sluggish workstation performance

06 Intermittent shared-resource access

## BUSINESS IMPACT

FINANCE + HUMAN RESOURCES

- Delayed access to shared documents
- Interrupted printing and routine tasks
- Reduced efficiency and productivity

*Even intermittent issues required investigation to protect service reliability.*

**OUR RESPONSE** Investigate • Prioritize • Correct where possible • Verify outcomes

# INVESTIGATION & CORRECTIVE ACTIONS

Our team reviewed system behavior, configuration settings, and test results to identify issues and verify corrective actions.

**INC-001** **SHARED FILE ACCESS**

Investigated connectivity, domain authentication, group membership, and folder permissions.

OUTCOME • Permissions corrected; authorized shared-folder access verified.

**INC-002** **SLOW DOMAIN LOGIN**

Reviewed NETLOGON events and domain-controller communication; restarted the Netlogon service.

OUTCOME • Team testing reported successful sign-ins in under five seconds.

**INC-003** **DESKTOP SETTINGS**

Reviewed Group Policy, reported SYSVOL / AD permissions concerns, and refreshed policies.

OUTCOME • Team testing reported restoration of expected desktop settings.

**THREE INCIDENTS • TECHNICAL FINDINGS AND VERIFIED / TEAM-REPORTED OUTCOMES**

EXECUTIVE SUMMARY

# INVESTIGATION & CORRECTIVE ACTIONS

Post-migration incident response | INC-004 – INC-006

## INC-004 NETWORK PRINTER AVAILABILITY

RESTORED • TEAM-REPORTED

Reviewed printer services and sharing configuration, then reconnected the missing printer. Windows Control Panel confirmed its return; the team reported a successful test print.

## INC-005 SYSTEM PERFORMANCE

FURTHER TESTING REQUIRED

Investigated resource usage and disabled EnterpriseMaintenanceTask, but the slowdown persisted. HDD-based VM storage was identified as a possible factor; disk optimization remains unverified.

## INC-006 INTERMITTENT SHARED FILE ACCESS

ACCESS RESTORED • TEAM-REPORTED

Tested connectivity, DNS, and access permissions. The team reported that correcting SharedData NTFS permissions restored authorized access during subsequent testing.

OUTCOME • Printer and file-access services restored in team testing; INC-005 remains under investigation.

# CURRENT SYSTEM STATUS & STABILITY

**POST-MIGRATION ASSESSMENT • CHARLIE'S ANGELS**

## RESTORED CORE SERVICES

✓ **Shared file access**
Authorized users regained access to shared resources.

✓ **Domain sign-in & desktop settings**
Login performance improved; expected policies reapplied.

✓ **Printer & network connectivity**
Network printer reconnected; server communication verified.

## OPEN ITEMS & RISKS

### INC-005 — Performance

Slowdown persisted after disabling the maintenance task. HDD storage remains a possible factor; further testing is required.

### DNS diagnostic concern

Anomalies observed, but not established as the cause of file-access problems.

**OVERALL: Key services restored during testing; continued monitoring and follow-up required before full stabilization.**

# LESSONS LEARNED & RECOMMENDATIONS

EXECUTIVE SUMMARY • CONTINUOUS IMPROVEMENT

## WHAT WE LEARNED

01 **Troubleshoot systematically**

Similar symptoms can have different causes after a migration.

02 **Verify, don’t assume**

A completed change is not proof that the user’s problem is resolved.

03 **Document and collaborate**

Shared findings and accurate records strengthen incident response.

## RECOMMENDED NEXT STEPS

01 **Continue performance monitoring**

Retest INC-005 and evaluate storage-related bottlenecks.

02 **Review permissions and domain services**

Periodically validate shared-folder access, DNS, and authentication.

03 **Test core business functions**

Check network access, desktop policies, and printer availability.

04 **Keep configuration records current**

Document changes, test outcomes, and any remaining open issues.

**KEY TAKEAWAY** **Reliable recovery depends on evidence-based fixes, verification, and continued follow-up.**

PLAN
TROUBLESHOOT
COLLABORATE
SOLVE
STABILIZE ♡
CHARLIE'S ANGELS
CAPSTONE PROJECT
PER SCHOLAS
Teamwork
Problem Solving
Real World Skills
Mission Complete! ♡
Auree-Lynn ♡
Christina ♡
Charles ♡
Aijahnae ♡
John ♡
NETWORKING
SYSTEM ADMIN
TROUBLESHOOTING
CYBERSECURITY
CLOUD
TEAMWORK
GOOD IDEAS BETTER SOLUTIONS ♡
Same Team Bigger Solutions ♡